Making a CPM of Collective Migration

Report on Assignment 1C

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1 Introduction

Collective cell migration concerns the movement of a group of cells in response to environmental or chemical cues, and is an emergent property of many interacting individual cells. Collective migration can happen when the volume of cells is enforced, but not too strongly so (λ Volume parameter). Cells further back can place new pixels inside a cell in the front on its backside, making the front cell 'under volume' and making it place new pixels on its other side more quickly. Through this interaction, the cell in the back can push the front cell to move more quickly than it otherwise would have. By using the Cellular Potts Model, we can explore how individual cells interacting with their environment give rise to such collective behavior.

Obstacles represent some environmental constraints that affect collective cell migration by guiding the movement of cells or disrupting it. By varying the number of obstacles, changes in the collective cell migration can be observed. Investigating these effects helps reveal the underlying mechanisms that guide cell and group behaviors. This experiment aims to explore how obstacles change the collective migration behaviour of cells by observing simulations with 4, 9, 16, or 25 obstacles.

2 Methodology

The Artistoo framework was used to implement the two-dimensional Cellular Potts Model simulations. The model was defined on a square grid of 200x200 pixels with wrapped dimensions. In addition to the background, there were two types of cells in the simulations: moving cells and obstacles. Obstacles were implemented by creating a new type of cell with different parameters in order to distinguish the two types of cells and to give them desired properties.

Obstacles were initialized with a target volume (V) of 150 pixels and volume constraint strength (λ_V) of 500 to ensure they are static and do not deform. Their perimeter (P) was set to 10 with a strong perimeter constraint strength (λ_P) of 200, keeping their shape strongly constrained. To further prevent any movement, their act ($\lambda_{\text{act}}, \text{Max}_{\text{act}}$) was set to 0. Additionally, the adhesion between obstacles and moving cells ($J_{\text{obstacle-cell}}$) was set to 1000 to ensure moving cells do not stick to obstacles or penetrate them.

On the other hand, moving cells had completely different parameters that would allow them to move and explore their environment, reproducing collective cell migration. These parameters were directly taken from the self-study exercise 1.3. In total, there were 37 moving cells on the grid. 37 was chosen because it is the maximum number of cells that allowed all obstacle configurations without breaking the simulation. Each moving cell was initialized with V of 200 pixels, making them (very roughly) twice the size of the obstacles, and λV of 50 to allow them moderate compressibility. Their P was set to 180 and λP to 2 to allow them to take on irregular shapes during their migration. Their movement, and the active migration, was controlled using activity constraints with λact of 200 and Maxact 20. These parameters allowed the reproduction of collective migration.

Table 1: CPM parameters for moving cells and obstacles. Shared parameters are listed in the "Shared Value" column.

Parameter	**Moving Cells**	**Obstacles**
Temperature (T)	20	20
Adhesion cell–matrix	20	20
Adhesion cell–cell	0	0
Adhesion obstacle–cell	1000	1000
Target volume (V)	200	100
Volume constraint (λ_V)	50	500
Target perimeter (P)	180	50
Perimeter constraint (λ_P)	2	200
Activity strength (λ_{act})	200	0
Activity memory (Max_{act})	20	0
Number of cells	40	9, 25, 49, 81
Obstacle radius	–	6
Obstacle padding	–	25

3 Results

4 Conclusion

5 Discussion

Appendices